

Todd Rylaarsdam

Full-Stack and Embedded Systems Engineer

St. Paul, MN 55014

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SKILLS

Software Native & Embedded C/C++ (STM32, Atmel, Nordic), Low Energy Bluetooth, TypeScript/NodeJS, Swift, Python, Databases (SQL, Firebase, InfluxDB), CI/CD, Google Cloud Platform, Tensorflow

Hardware PCB Design (Altium), Digital Logic Design (Verilog), 3D Modeling & Rendering (Solidworks), PCB Assembly Coordination, Hardware Debugging (SWD, Digital Logic Analyzers, Oscilloscopes)

PROFESSIONAL EXPERIENCE

MICROSOFT

Redmond, WA

Software Engineer Intern (Excel, 2024 & 2025)

May 2025–July 2025

May 2024–July 2024

Delivered multiple production ready, scalable, and tested solutions to improve and eliminate scenarios experienced millions of times per month that customers provided negative feedback related to Excel coauthoring. These solutions decreased the number of forced user reloads related to the use of Excel sheet views by over 73% and provided all users with a much more painless process to resume editing their workbook as normal after experiencing file corruption.

MAYO CLINIC

Rochester, MN

Undergraduate Intern - Neurology

May 2023–August 2023

Designed, created, and deployed to patient use a system to allow EEG implants to interface with an at-edge Tiny-ML capable microcontroller, enabling 24/7 real-time seizure detection and forecasting for patients with epilepsy, whether they were in the hospital, at home, or off-grid and disconnected from the internet.

Skills: C++, JavaScript, TensorFlow, Python, C#, 3D Modeling, Google Cloud, Grafana, InfluxDB, PCB Design

WINDY CITY LAB

Chicago, IL

Full Stack Embedded Systems Intern

May 2021–July 2022

Worked on projects for clients ranging from early stage IoT prototypes to production-ready sensors for scientific research. Responsibilities included communicating with clients like Argonne National Lab, Mayo Clinic and Northwestern University.

Developed a cloud-synchronized, distributed CPU emulator. Enabled summer students during the 2020 lockdowns to collaborate remotely, each building their own 8-bit CPU components. Once connected, these individual components formed a complete, distributed, and fully-functional CPU through cloud coordination.

Skills: C/C++, Node.JS, Python, PCB Design, 3D Modeling, Google Cloud, Swift, Kotlin

EDUCATION

GEORGE FOX UNIVERSITY

Newberg, OR

Bachelor of Science in Computer Science, Sophomore

August 2022–Present

GPA: 3.98, In-Major GPA: 4.00

AWARDS

- Meritorious Achievement Award in Sound Design, Kennedy Center American College Theater Festival, Region 7 – March 2024
- Oregon Site Champion (D2), International Collegiate Programming Contest – February 2024
- Ethel Ankeny Award for Technical Theater George Fox University – April 2023